

Global Navigation Satellite Systems With CENTIMETRE-LEVEL ACCURACY

Precision is designed, not accidental. Warehouse robots and drones require centimetre-level accuracy beyond the capability of standard global navigation satellite systems. Achieving this requires the right positioning technology, antenna design, sensor fusion, and verification. Electronics must be engineered to deliver reliable positioning in challenging environments. So how is it done?

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Imagine a robot moving through warehouse aisles, picking the required items from among many others lying in storage racks. Or a drone flying through a dense forest, reaching every checkpoint within centimetres. Such tasks are usually performed with the help of high-precision global navigation satellite system (GNSS) technology.

For R&D engineers working on such projects, the key question is not “Can we do it?” but “How do we design for consistent accuracy?” Standard GNSS, like that used in phones or cars, is not enough. Signals can bounce, drift, or get blocked. The atmosphere and hardware can also introduce errors.

The environment also affects accuracy. In cities, signals reflect off tall buildings, leading to positioning errors. In forests, trees block signals, reducing the reliability of positioning.

Precision positioning

Conventional GNSS, like the one in a smartphone, offers 5-10 metres of accuracy. This is generally sufficient for navigation and route planning. However, many applications require much higher precision down to 10cm or even 1cm.

Autonomous vehicles, for instance, need accurate positioning to change lanes safely and avoid obstacles. Even a 0.5-metre error can cause serious problems. In precision

agriculture, tractors spread fertiliser with centimetre-level accuracy to reduce waste. GNSS with $\pm 2.5\text{cm}$ accuracy can guide automated steering in large farms.

Surveying drones and inspection robots also rely on precise positioning. Errors of a few metres can disrupt missions and produce unreliable results. Industrial autonomous systems in warehouses and factories need centimetre accuracy to operate safely alongside people and machines.

Centimetre-level accuracy in such instances is a great boon, but we should also try to understand why it is difficult to achieve it. Standard GNSS performance can be affected by several sources of error (see table).

Possible solutions

Improving beyond the 5-10-metre accuracy of standard GNSS is not just about better hardware; it requires asking the right questions. Which correction method fits the application? How is the system designed? Which errors must the signal processing handle?

Centimetre-level accuracy requires correcting common errors. Atmospheric delays can slow signals. Satellite clock errors and small orbit inaccuracies also affect results. Signals may reflect off buildings or the ground before reaching the receiver. Each error seems minor, but together they significantly reduce accuracy.

The primary challenge lies in selecting the most suitable correction approach. Design engineers must balance infrastructure needs, convergence time, scalability, system complexity, and cost.

Solution A. Precision positioning solutions. Centimetre-level

SOURCES OF ERROR

Error Source	Why It Matters
Satellite clock and orbit errors	Satellites are not perfect, and orbit information has small inaccuracies
Ionospheric and tropospheric delays	Signals slow down and shift as they pass through the atmosphere layers
Receiver and antenna biases	The hardware itself introduces minor errors
Multipath effects	Signals bounce off buildings or other objects, confusing the receiver